

Quantitative Methods in Finance

Project: An empirical study looking at macroeconomic and financial variables' effects on the monthly returns of the S&P 500.

Contents

Maths part 1	3
Maths part 2	6
Abstract	9
Introduction	9
Methodology.....	12
Evaluation	26
Conclusion.....	30
Bibliography	32
Appendix.....	37

Maths part 1

1.1

Firstly, to define constrained optimization; it is the process of optimizing variable y , based on the variables of, with certain constraints. In this question we are concerned with finding the minimal variance of y , given a weighted mix of assets A, B and C, where the sum of the weights can not be greater than 1. The first order conditions are that the partial derivative must equal zero, as this is the point where the gradient is equal to zero, signifying a maximum or minimum point. The second condition is that summing the weights of each asset must equal 1

The process of optimization is as follows:

Step 1) Set-up the Lagrange function, substituting known coefficients and combining the constraint

Step 2) Calculate the partial derivatives of the Lagrange function and equal them to zero. This should be done for each variable.

Step 3) Create a matrix form, inverse the matrix and solve for the variables using matrix methods.

Step 4) Interpret the results

1.2

The purpose of this function is to add a constraint for weights of each asset A, B and C when optimizing the portfolio for minimum variance. The function states that the sum of the weights must not exceed 1. This indicates that we are not using leverage in the portfolio. The function is also needed because rearranged, it provides the weight of asset C which will be substituted into the Lagrange function.

1.3

(The function defining minimum portfolio variance, with the given constraint)

$$\text{Min}_{w_A, w_B, w_C} \sigma^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + w_C^2 \sigma_C^2 + 2w_A w_B \sigma_{AB} + 2w_A w_C \sigma_{AC} + 2w_B w_C \sigma_{BC}$$

$$\text{Constrained to } w_A + w_B + w_C = 1$$

(Substituting values to form Lagrange Function)

$$\begin{aligned} L(w_A, w_B, w_C, \lambda_1) \\ = 9w_A^2 + 16w_B^2 + 16w_C^2 + 6w_A w_B + 6w_A w_C - 8w_B w_C - \lambda_1(w_A + w_B + w_C - 1) \end{aligned}$$

(Taking partial derivatives and equating to zero for max/min)

$$\frac{\partial L}{\partial w_A} = 18w_A + 6w_B + 6w_C - \lambda_1 = 0$$

$$\frac{\partial L}{\partial w_B} = 32w_B + 6w_A - 8w_C - \lambda_1 = 0$$

$$\frac{\partial L}{\partial w_C} = 32w_C + 6w_A - 8w_B - \lambda_1 = 0$$

$$\frac{\partial L}{\partial \lambda_1} = -(w_A + w_B + w_C - 1) = 0$$

(Creating matrix form for the partial derivatives)

$$\begin{pmatrix} 18 & 6 & 6 & -1 \\ 6 & 32 & -8 & -1 \\ 6 & -8 & 32 & -1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} w_A \\ w_B \\ w_C \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

(Solve for variables, taking the inverse of matrix A and multiplying by vector B using R)

(See appendix 24 for code and 25 for solutions)

Optimal asset weights for minimal variance are as follows:

$$w_A = 0.333, w_B = 0.333, w_C = 0.333$$

This result implies that the three assets should be equally weighted within the portfolio. This distribution provides the portfolio with minimum variance. The Lagrange multiplier represents the shadow price of relaxing the constraint.

1.4

The difference between question 1.3 and 1.4 is that 1.4 has an additional constraint that is, the expected return of the portfolio denoted as $E(R_p)$. The portfolio should have an expected return of 18%. Thus, the weights of each asset must be modified dependent on each individual assets' $E(R)$, so the portfolio has the best possible risk/return profile.

(The function defining minimum portfolio variance, with the given constraints)

$$\text{Min}_{w_A, w_B, w_C} \sigma^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + w_C^2 \sigma_C^2 + 2w_A w_B \sigma_{AB} + 2w_A w_C \sigma_{AC} + 2w_B w_C \sigma_{BC}$$

$$\text{Constrained to } w_A + w_B + w_C = 1 \text{ and } E(R_p) = w_A E(R_A) + w_B E(R_B) + w_C E(R_C)$$

(Substituting values and constraints to form Lagrange Function)

$$\begin{aligned} L(w_A, w_B, w_C, \lambda_1, \lambda_2) \\ &= 9w_A^2 + 16w_B^2 + 16w_C^2 + 6w_A w_B + 6w_A w_C - 8w_B w_C \\ &\quad - \lambda_1(w_A + w_B + w_C - 1) - \lambda_2(10w_A + 20w_B + 20w_C - 18) \end{aligned}$$

(Taking partial derivatives and equating to zero for max/min)

$$\frac{\partial L}{\partial w_A} = 18w_A + 6w_B + 6w_C - \lambda_1 - 10\lambda_2 = 0$$

$$\frac{\partial L}{\partial w_B} = 32w_B + 6w_A - 8w_C - \lambda_1 - 20\lambda_2 = 0$$

$$\frac{\partial L}{\partial w_C} = 32w_C + 6w_A - 8w_B - \lambda_1 - 20\lambda_2 = 0$$

$$\frac{\partial L}{\partial \lambda_1} = -(w_A + w_B + w_C - 1) = 0$$

$$\frac{\partial L}{\partial \lambda_2} = -(10w_A + 20w_B + 20w_C - 18) = 0$$

(Creating matrix form for the partial derivatives)

$$\begin{pmatrix} 18 & 6 & 6 & -1 & 10 \\ 6 & 32 & -8 & -1 & -20 \\ 6 & -8 & 32 & -1 & -20 \\ 1 & 1 & 1 & 0 & 0 \\ 10 & 20 & 20 & 0 & 0 \end{pmatrix} \begin{pmatrix} w_A \\ w_B \\ w_C \\ \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 18 \end{pmatrix}$$

(Solve for variables, taking the inverse of matrix A and multiplying by vector B using R)

(see appendix 26 & 27)

Optimal asset weights for the given $E(R_P)$:

$$w_A = 0.2, w_B = 0.4, w_C = 0.4$$

Maths part 2

2.1

1	0.914762870549652	1.85633029071557
1	-0.793940637931373	0.029151815773688
1	-0.995358989788233	-1.56239635011936
1	-1.56613973514982	-0.524958110945035
1	0.0245323691186392	0.290483335450461
1	-0.445265430996678	-0.176126633782181
1	-0.694331471909081	-0.476715960155113
1	0.83741679717848	-0.883054130719283
1	0.1890552117237	-0.348126222243284
1	1.26889172866105	-0.0864933914415091

The dimension of this matrix is 10 x 3

This matrix represents the predictors and the intercept point represented by '1'.

2.2

The dimension of Matrix y is 10 x 1

The matrix represents all values for y

2.3

Matrix $(X' \cdot X)^{-1}$ was found by multiplying the transposed matrix with the original matrix and then taking the inverse (Appendix 28). This step is a component of the normal equation, necessary to calculate the values of coefficients in OLS regression. The matrix is defined as:

$$\begin{pmatrix} 0.1052 & 0.0060 & 0.0238 \\ 0.0060 & 0.1607 & -0.0755 \\ 0.0238 & -0.0755 & 0.1771 \end{pmatrix}$$

2.4

Matrix $X'y$ is found by multiplying the transposed X matrix with the y matrix. The code for finding the product of these two is found in (Appendix 29). This matrix summarizes the relationship between the independent variables and dependent variable. $X'y$ is also a component of the normal equation.

The matrix of this product has a dimension of 3 x 1 and is defined as:

$$\begin{pmatrix} 12.559 \\ 35.047 \\ 35.366 \end{pmatrix}$$

2.5

Multiplying the matrices from part 2.3 And 2.4 forms the normal equation which is $\beta = (X' \cdot X)^{-1} * (X'Y)$. The normal equation is used to solve for the coefficients relating to an OLS linear regression. The solution to this function provides a matrix of dimension 3 x 1, defined as:

2.375
3.039
3.916

The results indicate that the y intercept point is 2.375, and the estimated X1 and X2 coefficients are 3.039 and 3.916 respectively when all other variables are held constant.

2.6

Using R to create a linear regression model using the `lm()` function in R yields identical results that were calculated in section 2.5 (Appendix 30). They are the same because the `lm` function applies the normal equation method, just like the matrix computation method applied above. Using a computer program to calculate these relationships is beneficial because it eliminates human error and can process large data sets very quickly and provide easily scalability and editing. This is opposed to the matrix equation method which would be limited to a level of complexity and be a timelier operation.

Additionally, the `lm` function can provide diagnostic testing of the model afterwards, whereas the matrix equation solely provides the coefficients.

Abstract

This study investigates the relationship between macroeconomic and financial variables and their respective impact on the returns of the S&P 500 index. An OLS linear regression model was developed using 144 monthly data points over a 12-year period and 7 explanatory variables to begin with. The final was found using a top-down approach, by specifying the model based on literature findings and variables' p-values. The final model's five variables include: Unemployment, Interest rate, Inflation, Gold, and Oil. Changes in the price of gold and oil were found to cater for the largest returns or losses for the returns of the S&P 500 index, followed by interest. The model explains circa 20% of the variation in S&P 500 returns, reflecting the complexity of stock market dynamics and number of factors that influence asset pricing. Whilst the model meets key Ordinary Least Squares (OLS) assumptions, issues such as autocorrelation, heteroscedasticity, and multicollinearity required adjustments. Corrections were applied to meet the assumptions of Ordinary Least Squares (OLS), though the correlation between the variables shows they are not wholly independent, biasing the inferred effects of singular variables if all others were to be held constant. The findings provide a useful framework for understanding how changes in macroeconomic and financial conditions could possibly influence S&P 500 returns. This information can help decision trading making when used in conjunction with other financial indicators and research.

Introduction

The focus of this empirical study is to identify macro-economic and commodity variables that affect the returns of the Standard and Poor 500 index (S&P500). The study will seek to create a multiple linear regression model, that will provide insights regarding; which selected variables have the greatest impacts on S&P 500 returns, and the effects of changes in these variables regarding price directionality of the same index. The aim is that the findings are be able to aide S&P 500 investment decisions based upon changes within the macroeconomic and financial wellbeing of the economy.

The S&P 500 is a stock index for the 500 most prominent, publicly listed companies in America. Therefore, the index is an active gauge for the United States' share market as a whole, and used as a benchmark for market performance (Kenton 2024). As the S&P 500 is a proxy for the market return of global, marketable securities, being knowledgeable of variables that affect the index and their impacts, would be beneficial to support investment decisions in the financial markets for both individual stocks but also trading the index. With the S&P 500 performance being influenced by individually weighted stocks contained within the index, it is in fact the returns of these assets that contribute to the overall returns. Thus, the macroeconomic environment is an important aspect for which companies can prosper or go bust upon. Related to the macroeconomic environment is systematic risk. Systematic risk is a blanket term for market wide risk that can not be diversified away within a portfolio. Research has found that changes in the macro economy will influence asset prices and thus the S&P500 (Chen 1986). A changing investment environment would result in investors expected returns changing. Thus, it is further conclusive that macroeconomic factors influence investor risk appetite which in turn affects asset prices and firm valuations.

Theoretically speaking, the Efficient Market Hypothesis (EMH) says that at any given time, all securities will be accurately priced and reflect all currently available information (Fama 1970). On this basis asset prices should fluctuate on the release of all new information to the market. This rationale is supported by research that has identified; news releases invariably affect asset price movements in either direction (Ball 1986). The basis for stock movement exhibiting EMH is a random walk, meaning at any time stock prices can move up or down relating to new market information (Malkiel 2003). Based on this theory it is inferred that the S&P 500 will react to new information releases to the market, and thus exhibit a positive or negative reaction.

It is widely accepted that the macroeconomic environment is a major catalyst for changes in asset pricing as quoted by (Gaspareniene 2021). Evidently, company performance is related to the health of an economy, thus it can be inferred that macroeconomic events are over-arching variables relating to individual asset performance in the global economy. Attempting to explain the S&P 500 returns, a mix of macroeconomic and commodity data points will be examined.

These variables include inflation, interest, credit spread, unemployment, oil and gold price, and the USD foreign exchange rate (Dahlman 2023, Branson 1981, Alesina 1995, Pastor 2013).

Research by Boyd summarises that rising unemployment in expansionary economic periods positively affects asset pricing. Inversely, it has negative connotations during contractions (Boyd 2005). Further, it has been identified that interest rates and inflation have a large causality with stock returns (Alzoubi 2022, Chen 1986). Alzoubi also writes that stock market performance deteriorates when inflation and interest rates are high. Side-stepping to the topic of investment, it is common that foreign investment uses interest rates as a deciding factor. Hence the supply/demand of the USD can further be an indicator for investment in U.S based securities (Branson 1981). Finally, the credit spread between 10-year Treasury Bonds and 3-month Treasury bills, provides further insight to the bullish or bearishness of investors participating in the economy as a whole. A gripe with the credit spread, is that it moves in sync with interest rates, which could be a contributor to overspecification within the model. Significant research on the relationship between the stock market and changes in oil prices identifies that oil price shocks relating to future scarcity would have negative impacts, whilst an expansionary economy would have an increasing oil price, reflective of productivity (Bai 2021). Another popular commodity for economic uncertainty is gold. Gold is often seen as a reliable asset during times of economic instability and used as a tool for hedging by some more sophisticated investors and investment houses. The price of gold principally increases when investor confidence in financial markets declines, as it serves a valuable alternative to traditional investments (Baur and Lucey 2010).

To summarise the scope of the study, the research will follow a defined methodology, using a top-down approach to specify a final linear regression model. The model should act as a tool to be used in-line with other investment research methods to aide financial decision making in the markets. Upon completion, the evaluation will conclusively comment on the accuracy and use cases for predicting returns of the S&P 500 index, as well as identify any short comings.

Methodology

1.1 Introduction to the model format – OLS assumptions

The ability to create, model and analyse the effects of macro and financial variables on the price of the S&P 500 index is wholly dependent on the type, accuracy and variables being used. For the complexity of global markets and economies, where a multitude of different explanatory variables will affect the pricing and subsequent returns of the S&P 500. In this study, a multiple linear regression model will be used to provide more accurate feedback when analysing a series of complex relationships (Blokhin 2024).

Outlining the assumptions of the OLS model, the study will assume that the relationship between the explanatory variables and single response variable is linear in nature. This assumption is later proved true for the final model.

Further, the sample population used in this study is a random time-series selection, consisting of the prior 12 years of financial data to date. The data sample also meets the requirements of the number of observations being greater than the number of independent variables to be analysed, providing an ample amount of data sets to create a reliable model. On this basis, the second OLS assumption is catered for.

The third OLS assumption states that the mean of the error terms should be equal to zero. This assumption holds true in the case of this study, as the mean of the residuals of the regression is equal to zero (Appendix 22).

OLS assumption four states that there should not be perfect collinearity between independent variables used within the model. This assumption builds on the idea that the greater the variation in the X variables, the better the model is at predicting the impacts of X on Y. Therefore, correlation between X variables will affect the specificity of each estimator's effects on the independent variable. Using financial data, there is a magnitude of correlation between variables, which can cause problems for the OLS estimators. The model will assume correlation doesn't exist and address this problem when evaluating the results.

Assumption five assumes that residual all exhibit equal variance. Additionally, auto-correlation should not be present in the model. When conducting model testing, these are two areas where data may need to be treated, in order to meet the assumptions.

The final assumption states that error terms should exhibit a normal distribution. As previously mentioned, this test will need to be carried out when completing model testing. Resting upon a final model, the Assumptions will need to be carried out, to ensure validity and accuracy of the OLS regression model.

When forming the model, the final model will be determined, using a general to specific method, eliminating all variables that are deemed to be non-significant, have significant correlation effects, or conflicting results from literature published research. This means beginning with a broad selection of variables within the model, and specifying the model by removing and completing model testing one step at a time. The functional form of the OLS multiple linear regression model is shown below:

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \beta_4x_4 + \beta_5x_5 + \beta_6x_6 + \beta_7x_7 + \varepsilon$$

Delving further into the model, the estimated returns of the S&P 500 are derived from estimators for each independent variable. These estimators, multiplied by the change affecting the variable will provide the estimated effect on the returns of the S&P 500 index. Furthermore, the error term captures all other unexplained variation of the dependent variable that occurs from variables existing outside of the model. The changes in each variable and their effects on the S&P 500 returns can be defined by the following function:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1x_1 + \hat{\beta}_2x_2 + \hat{\beta}_3x_3 + \hat{\beta}_4x_4 + \hat{\beta}_5x_5 + \hat{\beta}_6x_6 + \hat{\beta}_7x_7 + \hat{\varepsilon}$$

1.2 Selection of variables

The selection of relevant explanatory variables and the correct use of data plays a large role in ensuring the model is accurate. The choice of a variable can be ambiguous depending on aims and what one judges to be a significant explainer of the dependent variable. Extrapolating on the idea of data collection, using data that exhibits trends will create a model that explains the independent variable based on the drift of the explanatory variables rather than the perceived

changes that occur to the variable. Because this study is concerned with identifying how changes in the selected variables effect the S&P 500 returns, the data must not exhibit trends. Furthermore, the model is focused on identifying individual variables' effects, with all else being *ceteris paribus*. Therefore, the variables should be as independent as possible. However, using macroeconomic data, there is a lot of interaction between economic indicators, resulting in the variables not being wholly independent of each other. Lastly, to ensure the model is accurate, it shouldn't be over or underspecified. Overspecification will result in less precise estimates, with the magnitude of a variable's effects being understated. Doubly so, an underspecified model may have a tendency to have variable bias where the effects are overestimated. Taking into account the aforementioned points, the selection of explanatory variable's to be used in the model is reasoned below:

Unemployment rate

The unemployment rate, which measures the percentage of the labor force that is jobless but actively seeking employment, is a pivotal indicator of a nation's economic health. Research shows that its relationship with the S&P 500 is multifaceted. High unemployment rates often signal economic contraction, marked by reduced consumer spending, weakened demand for goods and services, and lower corporate revenues, which negatively impacts investor sentiment and puts downward pressure on financial markets (Hess and Frantz 2005). Conversely, low unemployment rates suggest a robust and expanding economy, characterized by higher disposable incomes, increased consumer spending, and enhanced corporate performance. This environment tends to bolster stock prices, reflecting optimism in the financial markets (Blanchard 1990). It is identified that the effects of unemployment on the S&P 500 can vary depending on the economic cycle. During expansionary periods, rising unemployment may have a subdued impact as broader economic growth offsets labor market weaknesses (Okun 1962). However, during recessions, heightened unemployment exacerbates fears of prolonged economic downturns, leading to a sharper decline in stock prices (Shimer 2005). It can therefore be concluded that unemployment serves as an important variable that helps to understand movements in the broader economy and thus it's impacts on the returns of the S&P 500 index.

FED interest rate

The Federal Reserve's interest rate, known as the federal funds rate, is a key economic tool used to influence inflation, employment, and economic growth. It is the rate at which banks lend to each other overnight, and changes to this rate have profound effects on the economy. Research indicates that raising the federal funds rate makes borrowing more expensive, slowing economic activity to reduce inflation, while lowering the rate encourages spending and investment to stimulate growth (Taylor 1993). This rate indirectly affects consumer loans, mortgages, and savings rates, influencing broader economic behavior (Bernanke 1992). Adjustments to the federal funds rate are made during Federal Open Market Committee (FOMC) meetings based on economic indicators such as inflation and unemployment, with the goal of balancing price stability with maximum employment (Mishkin 2007). Additionally, it serves as a cornerstone of U.S. monetary policy and has a significant impact on global financial markets, as changes in the rate can affect capital investment, currency valuations and thus stock markets (Clarida 1999).

Inflation

A key macroeconomic metric, inflation tracks the overall rise in price levels over time and exerts significant influence on the S&P 500. Research suggests that its impact is nuanced: moderate inflation often reflects a thriving economy, fostering increased corporate revenues and lifting stock prices (Fama 1981). Conversely, elevated inflation erodes consumer purchasing power and inflates operational costs for companies, potentially squeezing profit margins and dampening investor confidence (Bodie 1976). Historically, periods of high inflation have coincided with weaker stock market performance, as reduced spending and higher expenses weigh on valuations (Geske and Roll 1983). Moreover, central banks typically counter rising inflation with monetary tightening measures, such as increasing interest rates, which can discourage equity investments (Mishkin 1990). Thus, inflation plays a pivotal role in shaping S&P 500 returns by directly influencing economic and market dynamics.

Credit spread

Credit spread, commonly measured as the difference between yields on long-term government bonds (e.g., 10-year Treasury bonds) and short-term Treasury bills (e.g., 3-month bills), serves as an important indicator for market risk and economic expectations. Research suggests that this spread reflects the premium investors demand for holding longer-term securities relative to shorter-term, lower-risk investments (Merton 1974). Its fluctuations provide valuable insights into investor sentiment and the prevailing macroeconomic environment. A widening credit spread typically signals increased perceived risk in the economy, often arising during periods of economic uncertainty, financial stress, or declining corporate profitability (Fama and French 1989). Investors, wary of deteriorating market conditions, demand higher yields on longer-term securities to compensate for the additional risk, which leads to reduced investor confidence and a pullback in equity markets, contributing to a decline in stock prices, including the S&P 500 (Chen, Roll, and Ross 1986). Conversely, a narrowing credit spread indicates reduced risk, increased liquidity, and economic stability, reflecting a more favorable investment climate where investors are willing to accept lower yields on long-term securities (Harvey 1989). This positive sentiment encourages greater investment in equities, driving stock prices higher and fostering growth in broader market indices like the S&P 500.

Overall, credit spread acts as a barometer for economic health, financial market stability, and investor risk appetite. Its influence on equity markets, particularly the S&P 500, underscores its value as an explanatory variable for forecasting stock market performance. By capturing shifts in market dynamics and investor sentiment, credit spread provides a nuanced perspective on the interconnectedness of macroeconomic factors and financial markets.

Crude oil and gold futures price

Oil and gold are core staples of the global economy and have impacts on financial markets, where the volatility can influence security pricing. Volatility in the oil price directly affects producers and consumers through the shifts in supply and demand. This is due to the dependence of oil in the global economy (Pindyck 2003). Similar to oil, gold plays an important role in the market economy.

Gold is often used as a hedge against inflation and risk of the unknown. For this reason, large institutional investors hold gold as a part of risk mitigation (Kaufmann 1989). The relationship between stock markets and gold is inverse, when markets are deemed safe, gold prices fall, however when there is greater uncertainty, investors will sell off financial securities and move funds into gold investments (Baur 2012). Research on the relationship between the commodities and stock market found that if the oil price increases, stock prices of companies that are dependent on oil for their operations decrease (Gilbert 1984). Research by (Ciner 2001) found that the variance in oil prices affects stock index returns, whilst (Ross 2024) says there is little correlation between the stock market and oil prices. Similarly, it was found that gold prices have a significant influence on the financial market returns (Batten 2010). (Gokmenoglu 2015) concludes in their research by stating that gold has higher short-term and long-term effects than oil, on the returns of the S&P 500.

GBP/USD

Research conducted on the link between currency exchanges and stock market indices is paradoxical. It is found that there is a relationship in the short term between the USD and U.S stocks (Bahmani-Oskooee 2006). This is supported by further research finding causality between depreciation of currency and negative stock market returns (Ajayi 1996). A deeper dive into the topic finds that the effects of foreign exchange rates on stock prices is not consistent between countries and markets. Stock market returns based on the variance in currency exchange rates may be lagged (Amihud 1993), meaning effects are seen at a future date, not instantly. Further research has also identified that correlation between stock returns and currency is variable, where the Philippines has negative correlation but South Korea has positive correlation (Granger 2000). It can be concluded that the relationship between stock markets and currencies is complex, with many factors at play that can affect the foreign exchange rate. Research has not been able to conclude with certainty that there is a significant relationship between stock market returns and currency appreciation/depreciation.

1.3 Early-stage data transformations

Data was compiled from a range of sources. The macroeconomic data was collected from the Federal Reserve Economic Data (FRED), and is presented on a monthly basis over 12 months. Because interest rate, unemployment rate, inflation rate and credit spread are already measured in percentage form, no data transformation was required.

Financial data concerned with the S&P 500, oil, gold and foreign exchange were sourced from (Investing.com). The data was presented as 'month end price', and it was evident that there were trends in the data which could bias the model output by finding a relationship between trends, rather than the effects of returns. The trends in the data can be witnessed in (Figures 1-4) below. Removing trends from the data, involved taking the natural log of each month's data and calculating the rates of change between each month. Thus, the financial data used in the study is presented as the returns, of the logs, of the month ending price. The transformed data thus exhibit mean reversion and will provide a more accurate model estimate. The logged data points can be seen in (Figure 5-8).

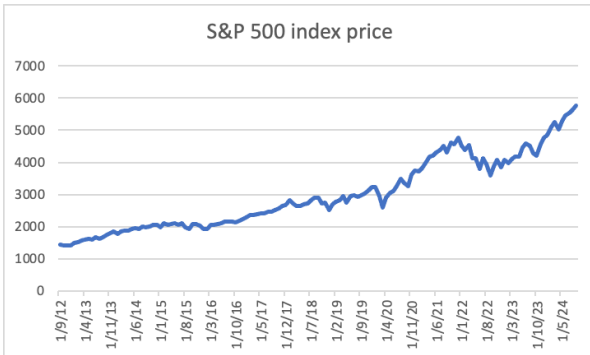


Figure 1. S&P 500 data

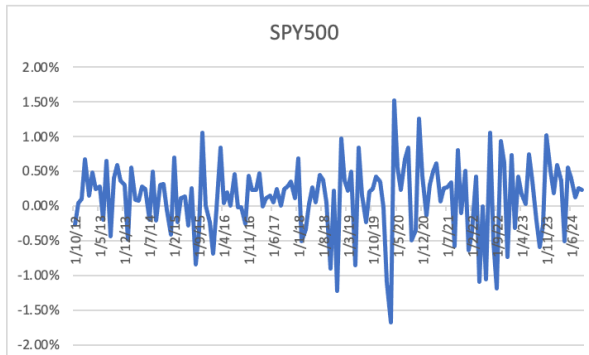


Figure 5. S&P 500 transformed data

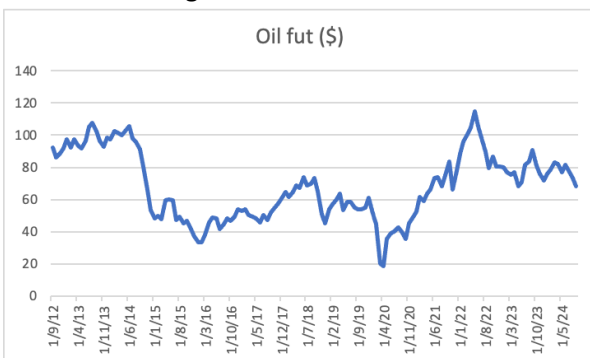


Figure 2. Oil futures price

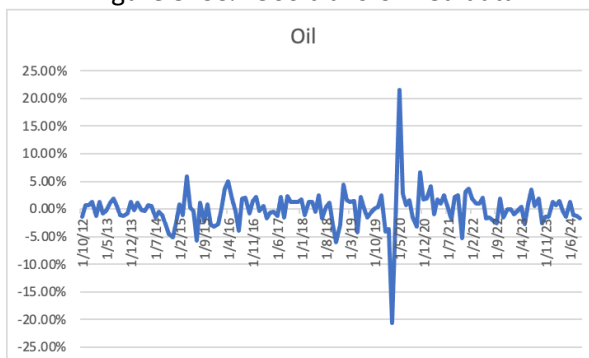


Figure 6. Oil futures transformed data

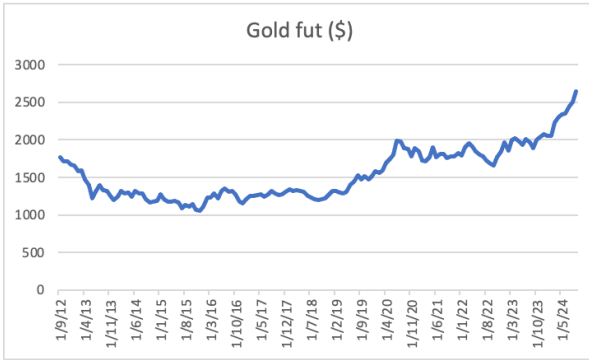


Figure 3. Gold futures price

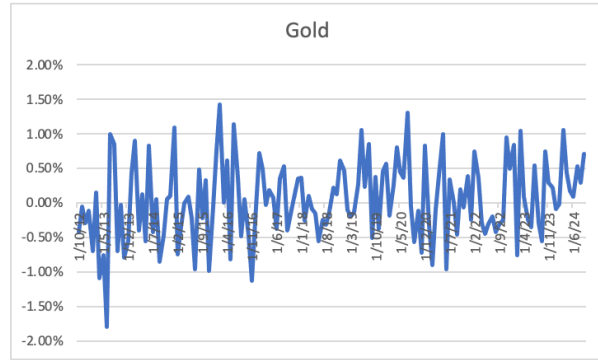


Figure 7. Gold futures transformed data



Figure 4. GBP/USD

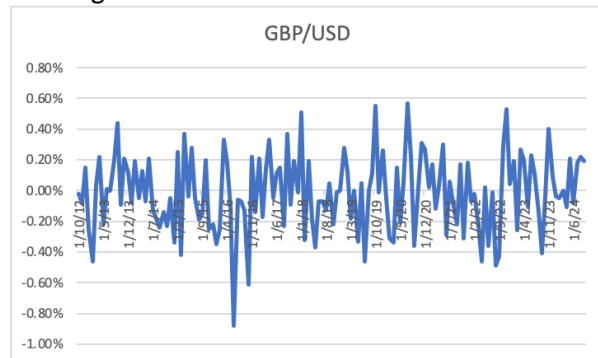


Figure 8. GBP/USD transformed data

1.4 Regression model

The method of approach, to create a model that can accurately predict S&P 500 returns follows the general to specific strategy. This practice focuses on removing one variable per testing cycle, with the intention to remove variables that are non-significant, exhibit correlation or their belonging is refuted by published journals. This is done in order to improve model accuracy. The use of macroeconomic data means that we can expect significant collinearity between variables stemming from the inter-relatedness cause and effects of these variables. Because they are deemed to not being wholly independent of each other, this will affect the estimated variables' coefficients. It will invoke that the estimated coefficient is not an independent effect of the variable, relating to the returns of the S&P 500. This conundrum can be countered by reducing the number of variables within the variable, reducing multi-collinearity and increasing the independence of each variable. When removing variables, caution needs to be taken, because, the removal of too many will result in an inaccurate model by overestimating the effects of each variable, affecting accuracy.

1.5 Testing the accuracy of the General model

Summary

The examined model has 144 data points, representing monthly data and returns over the past 12 years. The R-square value of the model is 0.37, indicating the model isn't over or underspecified (Appendix 1). The p-values for each dependent value showcase that unemployment rate, oil and the currency pair GBP/USD were significant predictors of the S&P 500 index (Appendix 2).

Normality test

The Jarque Bera test was conducted to test for a normal distribution of the residuals in the model. The null hypothesis being that the residuals follow a normal distribution is accepted; as the P value for this test was greater than 0.05 (Appendix 3). The model therefore meets the normality assumption of multiple linear regressions which states that when the dependent variable ($Y = \text{S\&P 500}$) is a linear function of independent variables ($X = \text{macroeconomic \& Financial variables}$) and the error terms, the regression is linear in parameters.

White test/Corrections

The White test was conducted to whether the variance of the errors in a regression model is constant (homoscedastic). The White test resulted in a test value of 24.39 and a p-value of 6.67E-05. As the p-value is well below the significance level of 0.05, the null hypothesis of homoscedastic variance is rejected. Exploring the OLS assumption of homoscedasticity, the error terms in the regression should all have the same variance. If this variance is not constant, we can conclude that the linear regression model has heteroscedastic tendencies and may yield inaccurate estimates. Because the model rejects the null hypothesis, it means the model exhibits heteroscedasticity. Data correction was conducted to remove the non-constant variance, using the coefficient test, resulting in new dependent variable p-values and error terms evidenced in (Appendix 4). Thus, the model now meets the OLS assumption of homogenous variance for all residuals.

Breusch Godfrey test

In this body of work, the Breusch-Godfrey test is used to detect autocorrelation in the residuals of a regression model. The results (Appendix 5) show that the Breusch-Godfrey test yields an LM test statistic value of 12.084 at 4 degrees of freedom and a p-value of 0.01674. As the p-value is smaller than the usual significance level of 0.05, the null hypothesis (no autocorrelation of the residuals) is rejected. This indicates there is autocorrelation in the residuals up to the fourth order, meaning past variable data does have an effect on future data points. This refutes the OLS assumption that error terms should not be correlated with lagged versions of themselves. A correction to the data was made to account for the autocorrelation, using the HAC approach (Appendix 6), providing new significance values and error terms for the independent variables.

Collinearity test

The OLS assumption of no multi-collinearity stipulates that you should have no linear relationships between the independent variables. In situations where there is an apparent linear relationship, it is best to remove a variable from the linear regression model. If the correlation between these variables is very strong, the OLS assumption states that the results will be biased and inaccurate, affecting the specificity of the model. To measure collinearity, the Variance Inflation Factor (VIF) for each variable is calculated. The factor shows how the different variables interact with each other. It is important that the independent variables have the least interaction between each other possible in order to attain an accurate idea of how each independent variable affects the S&P 500. For scale, a VIF of 1 indicates there is no correlation between variables, 1-5 suggests moderate collinearity and greater than 5 is significant (Frost 2017). The VIF values provided in (Appendix 7) highlight that interest rate, with a value of 7.35 is highly correlated to all the other variables, making it a less accurate explanatory variable. Further, the credit spread also has a high variance inflation factor of Values 5.59. These high values can be explained with reference to the similarity of the two variables, both being variables that have a high focus on interest rate movements. All other variables exhibit correlation with each other; however, it isn't to the same extent as the interest focused ones.

1.6.1 Model specification

Summary

In the process of specifying the model, the decision was made to remove the credit spread as an explanatory variable. This choice was primarily driven by concerns about potential collinearity between the credit spread and interest rates, which could distort the explanatory power of both variables in the model. The credit spread, often viewed as an indicator of broader long-term market health, is heavily influenced by the delta (change) in interest rates. This inherent dependency suggests that credit spread and interest rates are closely related, and including both in the model could lead to multicollinearity, which undermines the reliability of the individual coefficient estimates.

Moreover, statistical analysis revealed that the p-value for credit spread was insignificant, further supporting the decision to exclude it from the final model. An insignificant p-value implies that the credit spread does not provide meaningful additional explanatory power in predicting the S&P 500 returns, given the other variables already in the model.

After removing the credit spread, the model's summary output showed an R-squared value of 0.375. This is notably similar to the R-squared value obtained from the initial general model, reinforcing the validity of excluding the credit spread. This finding suggests that the overall explanatory power of the model remains relatively unchanged by the removal of the variable, confirming that its inclusion did not significantly enhance the model's ability to explain variations in the S&P 500 returns.

Additionally, the model maintained consistency in identifying the same significant explanatory variables (Unemployment, Interest, Inflation, Gold, and Oil), which further supports the robustness of the final specification. These variables continue to exhibit significant p-values (Appendix 8), underscoring their relevance in explaining stock market returns, even in the absence of the credit spread variable. This consistent output validates the decision to remove credit spread and strengthens the overall integrity of the model.

Normality test

A Jarque Bera test was run to test for normality of the residuals. The P value is greater than 0.05, indicating that the model's residuals follow a normal distribution, meeting the OLS assumption requirement for multiple linear regression models (Appendix 9).

White test/corrections

The White test provided a p-value less than 0.05, indicating that the model's residuals are heteroscedastic. The OLS assumption states that the variance in residuals should be constant, meaning it is homoscedastic. Therefore, the model was corrected to account for heteroscedasticity, providing new p-values and error terms for each variable (Appendix 10,11). This correction ensures that the model meets the OLS assumption of the data exhibiting a homoscedastic tendency.

Breusch Godfrey test

The Breusch Godfrey test found that there is autocorrelation of variables' residuals up to the fourth order, since the p-value was less than 0.05. This means that the error terms are influenced by lagged versions of themselves. Countering this finding, the model was adjusted using the HAC approach, which provided new p-values and standard errors for each explanatory variable (Appendix 12,13). The HAC test thus removes auto correlation that is present within the model, which meets the OLS assumption of the model not exhibiting auto correlation.

Collinearity test

The collinearity test was run to observe if there have been changes to the independence of each variable. It was found that the VIF of each variable was significantly reduced with the removal of credit spread. There is no longer significant correlation. The relationship between interest and inflation is evident, with the VIF showcasing the correlation between them. The results are a large improvement over the general models' and bodes well for analysing each variables' effects on S&P 500 returns, with all other factors being ceteris paribus (Appendix 14).

1.6.2 Further model specification

Summary

Further model specification involved removing the foreign exchange variable. Reasoning for this decision was that extensive literature review found that the effects of currency on the stock market returns is inconclusive. Further it was revealed that there is correlation between commodity price cycles and foreign exchange. This is significant, seeing as the U.S is a major oil producer. The new model summary provides an R-square of 0.2026 and has significant p-values for Unemployment, Interest and Oil (Appendix 15). The R square fell drastically over the previous model. The value of 0.2 is in the lower bounds of acceptance, and signifies that ~20% of the S&P 500 returns is explainable by the data set.

Normality test

Running the Jarque Bera test, the p-value is greater than 0.05. For this reason, we fail to reject the null hypothesis. This suggests that the data continues to follow a normal distribution as identified in the previous tests (Appendix 16). Thus, this result confirms that the OLS assumption of linear relationship continues to hold true for the model.

White test/corrections

Testing for homoscedasticity in the model, the white model provides a p-value that is less than 0.05. Based on this we reject the null hypothesis and accept that the variables' residuals exhibit variance across the population sample. Hence the model exhibits heterogeneity (Appendix 17). Corrections were made to account for the variation in residuals deviating from the fitted model, providing new error terms and p-values, evidenced in (Appendix 18). These corrections are necessary to meet the OLS assumption that the residuals' variance from the fitted values is homogenous.

Breusch Godfrey test

Running the test for autocorrelation, the p-value returned a value that is less than 0.05 (Appendix 19). The test concludes that there is autocorrelation within the data, signifying that past data influences future data. This result is consistent with the two previous model variations. Correcting for autocorrelation, the HAC approach was run, providing new error terms and p-values which account for the autocorrelation that is within the data (Appendix 20). Adjusting these values to account for autocorrelation means that the model has considered the effects of autocorrelation, and provided a new set of results that are adjusted, so the autocorrelation is removed. By doing so, the OLS assumption regarding autocorrelation is met.

Collinearity test

Undertaking the VIF test for the model, it was found that the removal of the foreign exchange variable reduced the collinearity for oil and gold. This decision was also associated with a small reduction in the VIF values of interest and inflation (Appendix 21). The collinearity has been significantly reduced as compared against the general model which was the starting point. This result strengthens the assumption that explanatory variables are independent of each other, providing more accurate estimates.

1.7 Final model

The final model incorporates five key variables: three macroeconomic factors—Unemployment, Interest rates, and Inflation—and two financial variables—Gold and Oil. The model is based on a dataset of 144 monthly data points, spanning a 12-year period, which captures historical movements in the S&P 500 returns. The model's R-squared value is 0.2026, indicating that 20% of the variation in the S&P 500 returns can be explained by the independent variables included. In the context of financial modeling, this R-squared value is considered acceptable, as stock market returns are influenced by a wide array of factors, many of which are difficult to capture in a single model.

Among the variables, Oil emerges as the most significant explanatory factor, with a highly significant p-value of 0.00037, which strongly indicates that oil prices play a critical role in explaining the variation in the S&P 500 returns. Following Oil, Unemployment and Inflation also prove to be significant explanatory variables, with p-values of 0.016 and 0.04, respectively (Appendix 20). These results suggest that macroeconomic factors, such as employment levels and inflation, continue to have meaningful effects on stock market performance.

The model also accounts for multicollinearity, ensuring that significant correlation between variables has been removed. While every effort was made to reduce collinearity, it is acknowledged that, given the combination of macroeconomic and financial data, a certain degree of correlation between variables remains inevitable. However, steps were taken to ensure that this correlation does not unduly influence the results, thereby improving the accuracy of the model's predictions and estimates.

A key decision in the modelling process was to retain all five variables in the final specification. Omitting relevant variables could result in an under-fitted model, which could lead to inaccurate estimates and unreliable statistical tests. By retaining all five variables, the model ensures that the relationships between S&P 500 returns and the selected variables are adequately captured, providing a more robust and reliable framework for understanding stock market behavior. This decision reflects the importance of including all relevant explanatory variables to avoid misspecification, which can distort results and lead to misleading conclusions.

Evaluation

The estimated model that estimates the relationship between the discussed explanatory variables and returns of the S&P 500 is:

$$\hat{y} = -0.00144 + 0.00064x_1 + 0.00076x_2 - 0.00058x_3 + 0.05118x_4 + 0.05475x_5 + \hat{\epsilon}$$

Where:

x_1 = Unemployment, x_2 = Interest, x_3 = Inflation, x_4 = Gold, x_5 = Oil

Analysing the coefficients of each variable, it can be concluded that unemployment has a positive impact on S&P returns if U.S unemployment increases. This statement is supported by (Boyd 2005) whom finds that “unemployment has positive effects during times of economic expansion”. Excessive and persistent unemployment can be linked to worse economic conditions in which case business’ may be struggling to sell their products and services and retain staff thus leading to a potential negative GDP growth and eventual recession. Thus, during recession periods, it can be deduced that unemployment has negative effects. This scenario was seen during the Corona Virus pandemic. It can be argued that the S&P 500 set new highs during this time, though this would be attributable to a low-interest rate environment and government stimulus that boosted the economy. The model indicates that interest rates have a positive relationship with stock market returns. That is as interest rates change by 1 unit, the expected percentage change of the S&P 500 is 0.00076%. This result contradicts the typically associated relationship between stock returns and interest rates- being that they are negatively correlated. A study by (Alam 2009) found that the correlation between interest rates and stock returns for 15 developed and developing countries is negative. If interest rates increase, investors are able to receive higher returns from bonds, hence sell down stock positions. Further, higher interest rates inhibit investment in projects. The positive coefficient for interest could be biased by the recent corona crash and economic recovery, subsequent high interest rates and the S&P 500 setting record highs. This period represents 4 years or 25% of the data. Related to interest, it was found that for every unit of change in inflation data, the S&P is predicted to change by -0.00058% . This figure is in line with financial theory that suggests inflation has negative impacts on stock returns because consumer spending falls (Zucchi 2023).

Looking at the returns of the two commodities, for each one percent change in the returns oil, the S&P 500 is expected to increase by 0.05118% . Similarly, the model forecasts that a one percent change in the returns of gold will cause a 0.05476% increase in the returns of the S&P 500. The positive coefficient for gold dissuades literature review, suggesting gold price increases when financial markets fall, or that gold is used as a hedge against the stock market. It is often stated that gold has a negative beta (Investopedia 2021). If this statement were to hold true, the coefficient for gold should be negative in this model. Commenting on the positive relationship

between oil and S&P 500 returns, it suggests that oil price increases have an overall positive effect on market returns.

The model is also supported by the Efficient Market Hypothesis (EMH). Semi-strong form EMH suggests that all publicly available information is already priced into publicly traded stocks and indexes. Any newly released information is reflected in asset pricing almost instantaneously. Based on this common financial theory, the model provides an estimate for the direction of movement for how the S&P 500 index will react, dependent upon the bullish or bearish narrative surrounding news releases. However, investors are forward thinking and expectations of macroeconomic data announcements may already be accounted for in asset pricing, reducing significance of price spikes. Speaking about the efficiencies of markets, human psychology can also bias investors' decisions. For this reason, the publication of new economic data could result in an over/under reaction in terms of market value- affecting the magnitude of returns. On this topic, for the General model, specified model and Further Specified model, the Breusch Godfrey test identified that there is autocorrelation within the data. This means that future returns are influenced by the dynamics of prior returns. For this reason, there is an argument that the variables and thus S&P 500 returns exhibit a degree of weak form efficiency. Weak form efficiency is based on the idea that future prices are dependent on historical price movements. This aspect was singularly treated, by applying a HAC correction to the model which provided new p-values and error terms. Thus, the model was adjusted to specifically identify the S&P 500 returns, based solely on changes in the variables, eliminating the effects of past returns on future returns.

Criticising the accuracy and dependency of the model, collinearity between independent variables exists. For this reason, the specificity of each variables' coefficient cannot be fully relied upon. In short, a single variable's effects on the S&P 500 returns is not 100% explanatory, with each of the other variables also influencing the effects. Additionally, the sample population of 144 months is small, meaning the results do not provide a true reflection. Within the data sample, there has been sustained periods of low interest rates (Figure 9) and a 'flash crash' in 2020 from the Corona virus outbreak (Figure 10). The crash was followed by hasty interest rate rises to curb inflation which also experienced a significant increase (Figure 11). For these reasons it can be

concluded that the sample size of 12 years and the abnormal macroeconomic environment of the past 4 years would bias the true effects of each variable relating to the returns of the S&P 500 index.

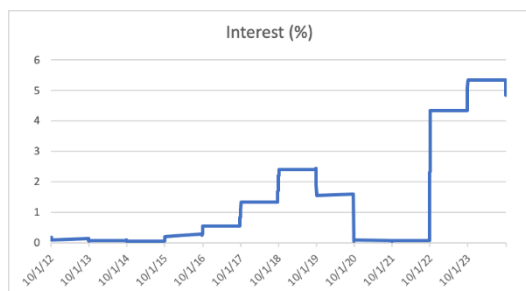


Figure 9. Interest rates over past 12 years

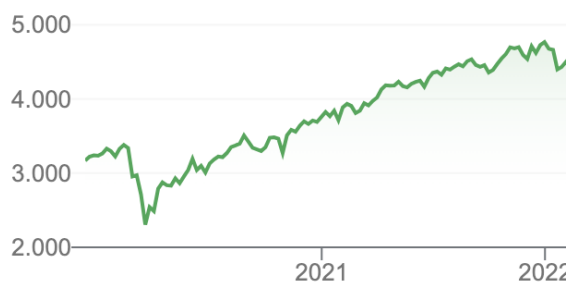


Figure 10. Corona crash in 2020

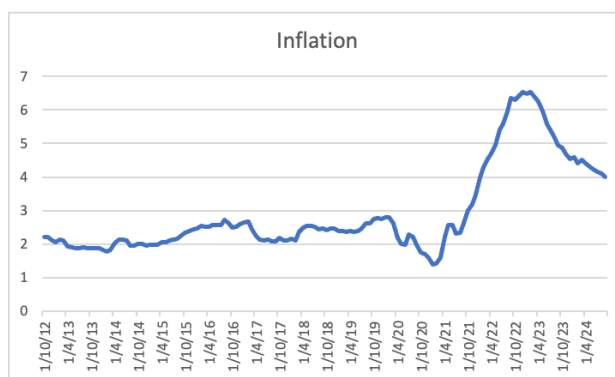


Figure 11. Inflation over past 12 years

The final specified model adheres to several OLS assumptions but also presents challenges. The normality assumption was met, as evidenced by the Jarque-Bera test results, which confirms the residuals follow a normal distribution. However, the model violates the homoscedasticity assumption, as the White test indicates there is heteroscedastic tendencies within the residuals. Corrections were applied to address this, ensuring homoscedastic error terms in subsequent results. Focusing on the residuals, the average value is equal to zero, further meeting OLS assumptions (Appendix 22). The model also exhibited autocorrelation, as identified by the results from Breusch-Godfrey test, which was corrected using the HAC approach to adjust for serial

correlation. Multicollinearity, was reduced through the removal of the foreign exchange rate variable, though it still persists to a milder extent, impacting the reliability of individual coefficients. This means the variables are not wholly independent. To summarise, the final model meets the required assumptions of the OLS regression which is supported by a linear trend line for the fitted vs residuals plot (Appendix 23).

Conclusion

The investigation surrounding the topic of identifying variables that influence S&P 500 returns has provided results that mostly align with common financial theory, but, there are outliers too. The effects of Interest and gold having positive correlation with the index's returns does not stem correctly, with all other variables being held constant. These specific findings of the model are widely refuted by large amounts of published literature on the topic.

With the data population sample being a monthly basis, across 12 years, the model looks at the longer term-effects of these studied variables on the returns, rather than on a day-to-day basis. For this reason, using the monthly period introduces a greater magnitude of uncertainty because as the time horizon increases, the accuracy and predictability diminish. However, the predicament is that much of the economic data is released on a monthly basis. Thus, it is also expected that there will be a large difference between the monthly S&P 500 performance explored in this study as opposed to analysing daily S&P 500 performance on the day of news releases.

This uncertainty is further supported by an abnormal macroeconomic time period as explained in the evaluation section of this report. A better data set would be to ignore data from 2020 and extend the time horizon backwards in order to generate a much larger sample population that could provide a more accurate model.

In conclusion, the model provides a basis for understanding what the effects of Unemployment, Interest, Inflation, Oil and Gold are on the returns of the S&P 500 index, with a monthly time frame. The model provides an estimate of the S&P 500 index returns based on changes in commodity prices or specific macroeconomic indicators. It is recommended to use this model as

a guide and use it in conjunction with other financial research methods and indicators, to paint a better picture of how the S&P may react to the variables explored in this study. This tool could be used to help generate profitable trades, by executing trades that profiteer off of directional movements in the markets, based on news releases and changes in pricing pertaining to the variables explored in this study.

Bibliography

Alesina, A., et al., n.d. Fiscal expansions and fiscal adjustments in OECD countries. Accessed: 17 October 2024.

Ajaji, R., et al., n.d. On the dynamic relationship between stock prices and exchange rates. Accessed: 19 November 2024.

Alam, M., et al., n.d. Relationship between interest rate and stock price: Empirical evidence from developed and developing countries. Accessed: 11 December 2024.

Amihud, Y., et al., n.d. Exchange rates and the valuation of equity shares. Accessed: 19 November 2024.

Alzoubi, M., 2022. Stock market performance: Reaction to interest rates and inflation rates. Accessed: 14 October 2024.

Bai, C., et al., n.d. The impact of oil prices on the U.S Stock Market. Accessed: 15 October 2024.

Ball, R., et al., 1986. An empirical evaluation of accounting income numbers. Accessed: 17 October 2024.

Batten, J., et al., n.d. The macroeconomic determinants of volatility in precious metals markets. Accessed: 21 November 2024.

Baur, D., et al., 2010. Is gold a hedge or a safe haven? An analysis of stocks, bonds and gold. Accessed: 24 October 2024.

Baur, D., et al., 2012. Asymmetric volatility in the gold market. Accessed: 21 November 2024.

Bahmani-Oskooee, M., et al., n.d. Stock prices and the effective exchange rate of the dollar. Accessed: 19 November 2024.

Blokhin, A., 2024. Linear Vs Multiple Regression: What's the difference. Accessed: 14 October 2024.

Boyd, J., et al., n.d. The stock Market's reaction to unemployment News: Why Bad News is usually good for stocks. Accessed: 15 October 2024.

Branson, W. H., n.d. Macroeconomic determinants of real exchange rate risk. Accessed: 17 October 2024.

Chen, N., et al., n.d. Economic Forces and the Stock Market. Accessed: 15 October 2024.

Ciner, C., n.d. Energy shocks and financial markets: Non-linear linkages. Accessed: 21 November 2024.

Dahlman, A., 2023. Macroeconomic variables and the stock market: A U.S. study of their relationship. Accessed: 14 October 2024.

Fama, E., 1970. Efficient Capital Markets: A Review of Theory and Empirical Work. Accessed: 14 October 2024.

Fama, E., 1981. Stock returns, real activity, inflation, and money. Accessed: 17 October 2024.

Frost, J., n.d. Multicollinearity in regression analysis: Problems, Detections & Solutions. Accessed: 20 November 2024.

FRED, n.d. Federal Reserve Economic Data of St Louis. Accessed: <https://fred.stlouisfed.org>

Gaspareniene, L., 2021. Modelling of S&P 500 Index Price Based on U.S. Economic Indicators: Machine Learning Approach. Accessed: 14 October 2024.

Gilbert, R., et al., n.d. Will oil markets tighten again? A survey of policies to manage possible oil supply disruptions. Accessed: 21 November 2024.

Gokmenoglu, K., et al., n.d. The interactions among gold, oil and stock market: Evidence from S&P 500. Accessed: 21 November 2024.

Granger, C., et al., n.d. A Bivariate Causality between Stock Prices and Exchange Rates.
Accessed: 19 November 2024.

Investing.com, n.d. Accessed: <https://www.investing.com>

Investopedia, n.d. Using Beta to understand a stock's risk. Accessed: 7 December 2024,
<https://www.investopedia.com/investing/beta-gauging-price-fluctuations>

Kaufmann, T., et al., n.d. The price of gold: A simple model. Accessed: 21 November 2024.

Kenton, W., 2024. S&P 500 Index: What is it and why is it important in investing. Accessed: 14 October 2024.

Malkiel, B., 2003. The Efficient Market Hypothesis and Its Critics. Accessed: 14 October 2024.

Pastor, L., et al., 2013. Political uncertainty and risk premia. Accessed: 17 October 2024.

Pindyck, R., n.d. Volatility in natural gas and oil markets. Accessed: 21 November 2024.

Ramelli, S., et al., 2020. Feverish stock price reactions to COVID-19. Accessed: 17 October 2024.

Ross, S., n.d. How oil prices affect stock returns. Accessed: 7 December 2024,
<https://www.investopedia.com/ask/answers/030415/how-does-price-oil-affect-stock-market.asp>

Zucchi, K., n.d. Inflation's impact on stock returns. Accessed: 7 December 2024,
<https://www.investopedia.com/articles/investing/052913/inflations-impact-stock-returns.asp>

Appendix

Regression Statistics

Multiple R	0.609646
R Square	0.371668
Adjusted R Square	0.339328
Standard Error	0.004195
Observations	144

Appendix 1. Regression statistics showcasing R^2 .

P-value

Intercept	0.774235331
Unemployment	0.028542296
Interest	0.507789695
Inflation	0.401667934
Credit spread	0.779590612
Oil	0.001631163
Gold	0.865569127
GBP/USD	1.30E-08

Appendix 2. P values of variables in the unspecified model

Jarque Bera Test

data: residuals

X-squared = 1.1143, df = 2, p-value = 0.5728

Appendix 3. Jarque-Bera test of unspecified model

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.00068292	0.00261743	-0.2609	0.79455
Unemployment	0.00053636	0.00031437	1.7061	0.09027 .
Interest	0.00034477	0.00052858	0.6522	0.51534
Inflation	-0.00033881	0.00050551	-0.6702	0.50384
Creditspread	-0.00019530	0.00069445	-0.2812	0.77897
Oil	0.03603654	0.01501857	2.3995	0.01777 *
Gold	-0.00972233	0.06115571	-0.1590	0.87392
GBPUSD	0.91114569	0.15892093	5.7333	6.088e-08 ***

Appendix 4. Applied coefficient test to data to remove heteroskedasticity- new variable coefficients

Breusch-Godfrey test for serial correlation of order up to 4

data: model

LM test = 12.084, df = 4, p-value = 0.01674

Appendix 5. Breusch-Godfrey test

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.00068292	0.00248805	-0.2745	0.78413
Unemployment	0.00053636	0.00026418	2.0302	0.04428 *
Interest	0.00034477	0.00046575	0.7402	0.46043
Inflation	-0.00033881	0.00027976	-1.2111	0.22797
Creditspread	-0.00019530	0.00058853	-0.3318	0.74052
Oil	0.03603654	0.01400412	2.5733	0.01115 *
Gold	-0.00972233	0.05666337	-0.1716	0.86402
GBPUSD	0.91114569	0.17516224	5.2017	7.099e-07 ***

Appendix 6. Autocorrelation corrections to coefficients using HAC method

```
> print(vif_values)
```

```
Unemployment    Interest    Inflation Creditspread    Oil    Gold
1.591528        7.354478    2.423509    5.595707    1.128105    1.084589
GBPUSD
1.123935
```

Appendix 7. Multicollinearity test

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.0011222	0.0018211	-0.616	0.53877
Unemployment	0.0005487	0.0002379	2.307	0.02257 *
Interest	0.0004660	0.0003147	1.481	0.14093
Inflation	-0.0003340	0.0004082	-0.818	0.41460
Oil	0.0361783	0.0111446	3.246	0.00147 **
Gold	-0.0071264	0.0636309	-0.112	0.91099
GBPUSD	0.9105068	0.1500651	6.067	1.2e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.004177 on 137 degrees of freedom

Multiple R-squared: 0.3715, Adjusted R-squared: 0.344

F-statistic: 13.5 on 6 and 137 DF, p-value: 5.42e-12

Appendix 8. Specified model summary output

Jarque Bera Test

data: residuals1

X-squared = 1.0995, df = 2, p-value = 0.5771

Appendix 9. Jarque-Bera test of specified model

```
> print(white1)
# A tibble: 1 × 5
  statistic p.value parameter method alternative
  <dbl>    <dbl>    <dbl> <chr>    <chr>
1    22.2  0.0354      12 White's Test greater
```

Appendix 10. White test of specified model

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.00112219	0.00261743	-0.4287	0.66879
Unemployment	0.00054872	0.00031437	1.7455	0.08315 .
Interest	0.00046597	0.00052858	0.8815	0.37957
Inflation	-0.00033401	0.00050551	-0.6608	0.50988
Oil	0.03617827	0.01501857	2.4089	0.01733 *
Gold	-0.00712645	0.06115571	-0.1165	0.90740
GBPUSD	0.91050681	0.15892093	5.7293	6.134e-08 ***

Appendix 11. Heteroscedasticity correction to specified model

Breusch-Godfrey test for serial correlation of order up to 4

data: modelsp1
LM test = 12.065, df = 4, p-value = 0.01687

Appendix 12. Breusch-Godfrey test for specified model

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.00112219	0.00248805	-0.4510	0.65268
Unemployment	0.00054872	0.00026418	2.0770	0.03967 *
Interest	0.00046597	0.00046575	1.0005	0.31885
Inflation	-0.00033401	0.00027976	-1.1939	0.23458
Oil	0.03617827	0.01400412	2.5834	0.01083 *
Gold	-0.00712645	0.05666337	-0.1258	0.90010
GBPUSD	0.91050681	0.17516224	5.1981	7.156e-07 ***

Appendix 13. Corrections to account for autocorrelation in specified model

```
> print(vif_values1)
```

Unemployment	Interest	Inflation	Oil	Gold	GBPUSD
1.540609	2.612587	2.419421	1.125878	1.062948	1.123686

Appendix 14. VIF values of specified model.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.0014432	0.0020429	-0.706	0.4811
Unemployment	0.0006421	0.0002664	2.410	0.0173 *
Interest	0.0007618	0.0003489	2.184	0.0307 *
Inflation	-0.0005810	0.0004558	-1.275	0.2045
Oil	0.0511768	0.0121958	4.196	4.83e-05 ***
Gold	0.0547548	0.0704885	0.777	0.4386

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.004688 on 138 degrees of freedom
Multiple R-squared: 0.2026, Adjusted R-squared: 0.1737
F-statistic: 7.013 on 5 and 138 DF, p-value: 7.201e-06

Appendix 15. Specified model V2 model summary

Jarque Bera Test

data: residuals2
X-squared = 2.7428, df = 2, p-value = 0.2538

Appendix 16. Specified model V2 normality test

```
> print(white2)
# A tibble: 1 × 5
  statistic p.value parameter method alternative
  <dbl>    <dbl>    <dbl> <chr>    <chr>
1    35.5 0.000103      10 White's Test greater
> |
```

Appendix 17. Specified model V2 white test

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-0.00144316	0.00261743	-0.5514	0.5822753	
Unemployment	0.00064212	0.00031437	2.0426	0.0430004	*
Interest	0.00076182	0.00052858	1.4413	0.1517776	
Inflation	-0.00058104	0.00050551	-1.1494	0.2523666	
Oil	0.05117678	0.01501857	3.4076	0.0008592	***
Gold	0.05475481	0.06115571	0.8953	0.3721674	

Appendix 18. Heteroscedasticity correction for specified model V2

Breusch-Godfrey test for serial correlation of order up to 4

data: modelsp2

LM test = 19.939, df = 4, p-value = 0.0005134

Appendix 19. Specified model V2 Breusch Godfrey test

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-0.00144316	0.00248805	-0.5800	0.5628358	
Unemployment	0.00064212	0.00026418	2.4306	0.0163586	*
Interest	0.00076182	0.00046575	1.6357	0.1041843	
Inflation	-0.00058104	0.00027976	-2.0769	0.0396647	*
Oil	0.05117678	0.01400412	3.6544	0.0003653	***
Gold	0.05475481	0.05666337	0.9663	0.3355757	

Appendix 20. Specified model V2 variables' corrections for autocorrelation

```
> print(vif_values2)
```

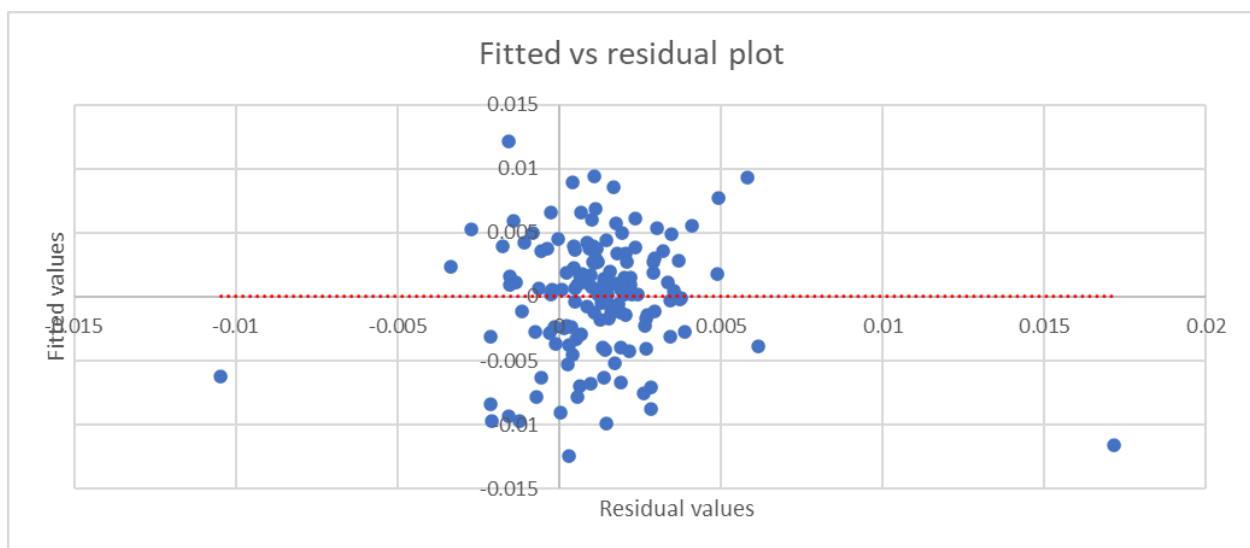
Unemployment	Interest	Inflation	Oil	Gold
1.534158	2.549845	2.395350	1.070485	1.035641

Appendix 21. Specified model V2 variables' VIF values

RESIDUAL OUTPUT

<i>Observation</i>	<i>Predicted SPY500</i>	<i>Residuals</i>	<i>Residual average</i>
1	0.001434	-0.00418	2.16E-13
2	0.002665	-0.00227	
3	0.002678	-0.00171	
4	0.003221	0.003551	
5	0.001339	0.000168	
...			
...			
...			

Appendix 22. Specified model Average of residuals



Appendix 23. Final model's linear trend line between fitted vs residual

```
# Define matrix A
A <- matrix(c(18, 6, 6, -1,
              6, 32, -8, -1,
              6, -8, 32, -1,
              1, 1, 1, 0),
            nrow = 4, byrow = TRUE)

# Define vector b
b <- c(0, 0, 0, 1)

# Inverse of A
A_inv <- solve(A)

# Multiply inverse of A by b
solution <- A_inv %*% b

# Show solution
solution
```

Appendix 24. R code to solve for portfolio weights

```
> solution
      [,1]
[1,] 0.3333333
[2,] 0.3333333
[3,] 0.3333333
[4,] 10.0000000
```

Appendix 25. Solutions for each asset weight and lambda

```
# Define matrix A
A <- matrix(c(18, 6, 6, -1, 10,
              6, 32, -8, -1, -20,
              6, -8, 32, -1, -20,
              1, 1, 1, 0, 0,
              10, 20, 20, 0, 0),
            nrow = 5, byrow = TRUE)

# Define vector b
b <- c(0, 0, 0, 1, 18)

# Inverse of A
A_inv <- solve(A)

# Multiply inverse of A by b
solution <- A_inv %*% b

# Show solution
solution
```

Appendix 26. R code to solve for portfolio weights, given two constraints

```
> solution
      [,1]
[1,] 0.20
[2,] 0.40
[3,] 0.40
[4,] 9.20
[5,] 0.08
```

Appendix 27. Solutions for each asset weight and lambda

```
# matrix X
X <- matrix(c(
  1, 0.914762871, 1.856330291,
  1, -0.793940638, 0.029151816,
  1, -0.99535899, -1.56239635,
  1, -1.566139735, -0.524958111,
  1, 0.024532369, 0.290483335,
  1, -0.445265431, -0.176126634,
  1, -0.694331472, -0.47671596,
  1, 0.837416797, -0.883054131,
  1, 0.189055212, -0.348126222,
  1, 1.268891729, -0.086493391
), nrow = 10, ncol = 3, byrow = TRUE)

# XT times X
xtX <- t(X) %*% X

# result
print(xtX_inv)
```

Appendix 28. R code for finding inverse matrix

```
# vector y
y <- c(12.31559705, 0.211112275, -6.459926294, -5.053962259, 4.015936736,
      0.43825592, -1.570646528, 1.09847194, 1.767314269, 5.796690889)

# Compute  $X^T \cdot y$ 
xt_y <- t(X) %*% y

# Print the result
print(xt_y)
```

Appendix 29. R code to find $X'y$

```
Call:
lm(formula = y ~ x1 + x2, data = Data)

Residuals:
    Min       1Q   Median       3Q      Max
-0.61488 -0.10655  0.06774  0.16864  0.42788

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.3759     0.1145   20.75 1.52e-07 ***
x1             3.0387     0.1415   21.48 1.20e-07 ***
x2             3.9163     0.1485   26.37 2.89e-08 ***
```

Appendix 30. Coefficients using R built in model function

```
47 summary(model)
48
49 #Create model
50
51 model <- lm(SPY500 ~ Unemployment + Interest + Inflation + Creditspread + Oil +
52
53 #Summary of model
54 summary(model)
55
56 #Test for normality
57 residuals <- residuals(model)
58 install.packages("tseries")
59 library(tseries)
60 jb_test <- jarque.bera.test(residuals)
61 print(jb_test)
62 P Value> 0.05, accept null hypothesis that the data follows a normal distributio
63
64 #White test for heteroskedasticity
65 library(lmtest)
66 model <- lm(SPY500 ~ Unemployment + Interest + Inflation + Creditspread + Oil +
67 white_test <- bptest(model, ~ fitted(model) + I(fitted(model)^2))
68 print(white_test)
69 library(lmtest)
70 white(model)
71 white <- white(model)
72 print(white)
73 P value<0.05, reject null hypothesis, data is heteroskedastic
74
75 #Apply correction to heteroskedastic data
76 hac_se <- vcovHAC(model)
77 hac_results <- coeftest(model, vcov = hac_se)
78 print(hac_results)
79
80 #Breusch Godfrey test
81 bg_test <- bgtest(model, order = 4)
82 print(bg_test)
83
```

Code for General model


```
#Create model

modelspl1 <- lm(SPY500 ~ Unemployment + Interest + Inflation + Oil + Gold + GBPU)
#Summary of model
summary(modelspl1)
#Test for norrmality
residuals1 <- residuals(modelspl1)
jb_test1 <- jarque.bera.test(residuals1)
print(jb_test1)
P Value> 0.05, accept null hypothesis that the data follows a normal distributi
#White test for heteroskedasticity
white_test1 <- bptest(modelspl1, ~ fitted(model) + I(fitted(model)^2))
print(white_test1)
library(lmtest)
white1 <- white.test(modelspl1)
print(white1)
P value<0.05, reject null hypothesis, data is heteroskedastic
#Apply correction to heteroskedastic data
coeftest(modelspl1, vcov = vcovHC(model, type = "HC"))
#Breusch Godfrey test
bg_test1 <- bgtest(modelspl1, order = 4)
print(bg_test1)
coeftest(modelspl1, vcov = vcovHAC(model))
#Multicollinearity
vif_values1 <- vif(modelspl1)
print(vif_values1)
```

Code for model v1

```
#Create model
modelspl2 <- lm(SPY500 ~ Unemployment + Interest + Inflation + Oil + Gold, data = 
#Summary of model
summary(modelspl2)
#Test for norrmality
residuals2 <- residuals(modelspl2)
jb_test2 <- jarque.bera.test(residuals2)
print(jb_test2)
P Value> 0.05, accept null hypothesis that the data follows a normal distributio
#White test for heteroskedasticity
white2 <- white(modelspl2)
print(white2)
P value<0.05, reject null hypothesis, data is heteroskedastic
#Apply correction to heteroskedastic data
coeftest(modelspl2, vcov = vcovHC(model, type = "HC"))
#Breusch Godfrey test
bg_test2 <- bgtest(modelspl2, order = 4)
print(bg_test2)
coeftest(modelspl2, vcov = vcovHAC(model))
#Multicollinearity
vif_values2 <- vif(modelspl2)
print(vif_values2)
summary(modelspl2)
```

Code for final model